# CONSIDERATIONS WHEN SPECIFYING A TRANSFORMER

- Application
- Configuration
- Internal Construction
- Dielectric Fluid
- Accessories

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# APPLICATION

- Distribution – Running lights, HVAC, Office
- Motor Control Center
- Dedicated Load – Pump, Crusher
- Extreme Duty – Drag-line

# Available Configurations

- Padmount
- Station
- Secondary Unit Substation
- Specialty
  - Grounding, Rectifier, Mine Duty

# INTERNAL CONSTRUCTION

**Shell vs. Core Form Transformers**

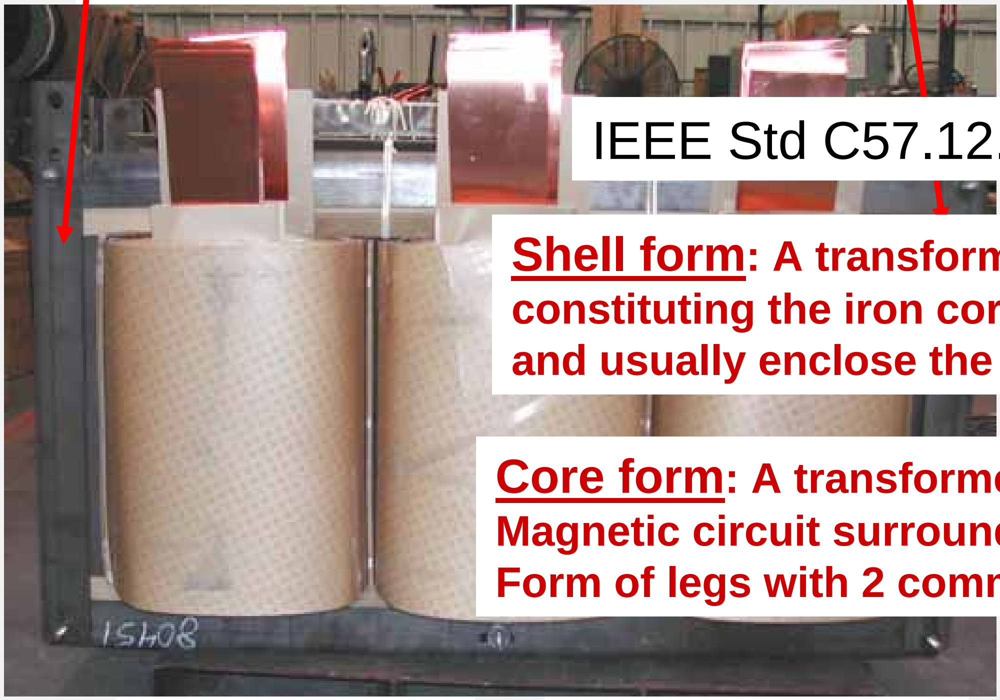
**Rectangular vs. Round Coil Construction**

**Disk Wound**

# Shell form & core form transformer construction

-5 legged wound core-
Each phase has a magnetic flux return path independent from the other phases.

- 3 legged stacked core-
Each phase shares a core bar. Harmonic content can lead to tank heating if wye/wye connected.



IEEE Std C57.12.80-2002 definitions:

**Shell form: A transformer in which the laminations constituting the iron core surround the windings and usually enclose the greater part of them.**

**Core form: A transformer in which those parts of the Magnetic circuit surrounded by the windings have the Form of legs with 2 common yokes.**

# Shell & core transformers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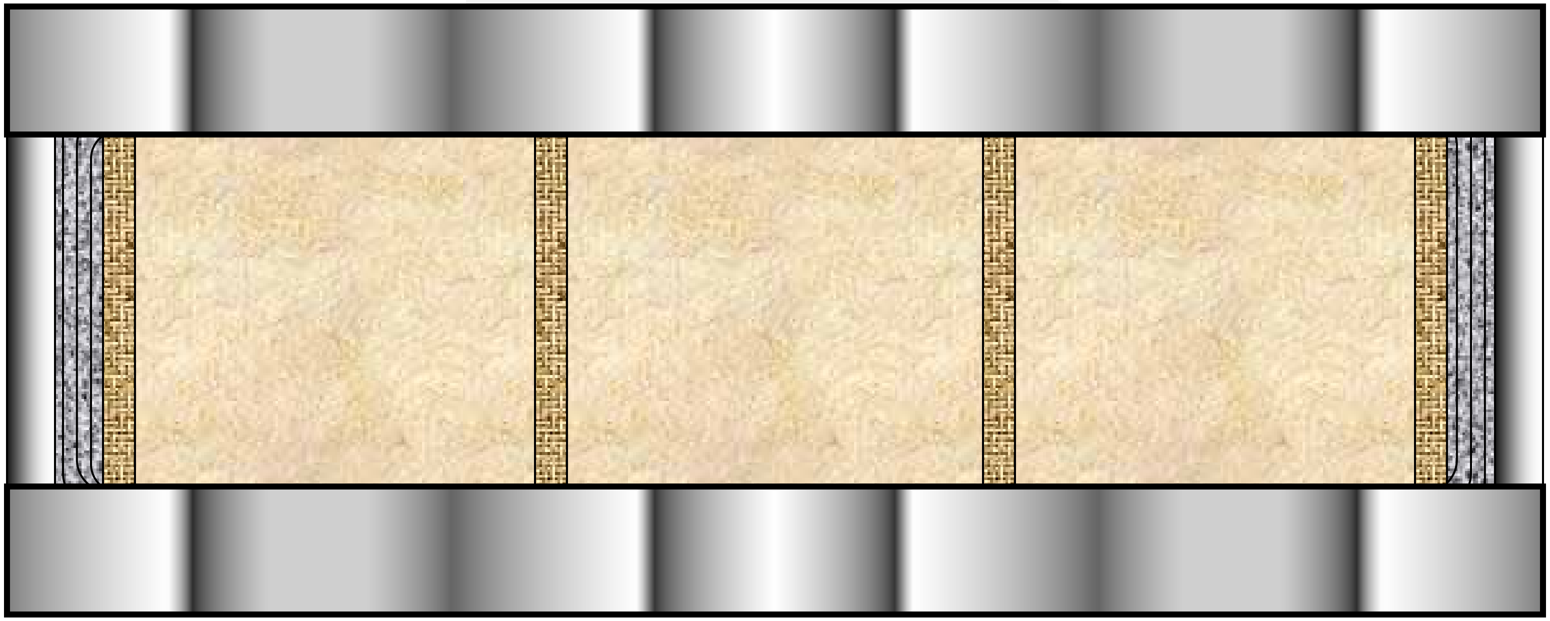
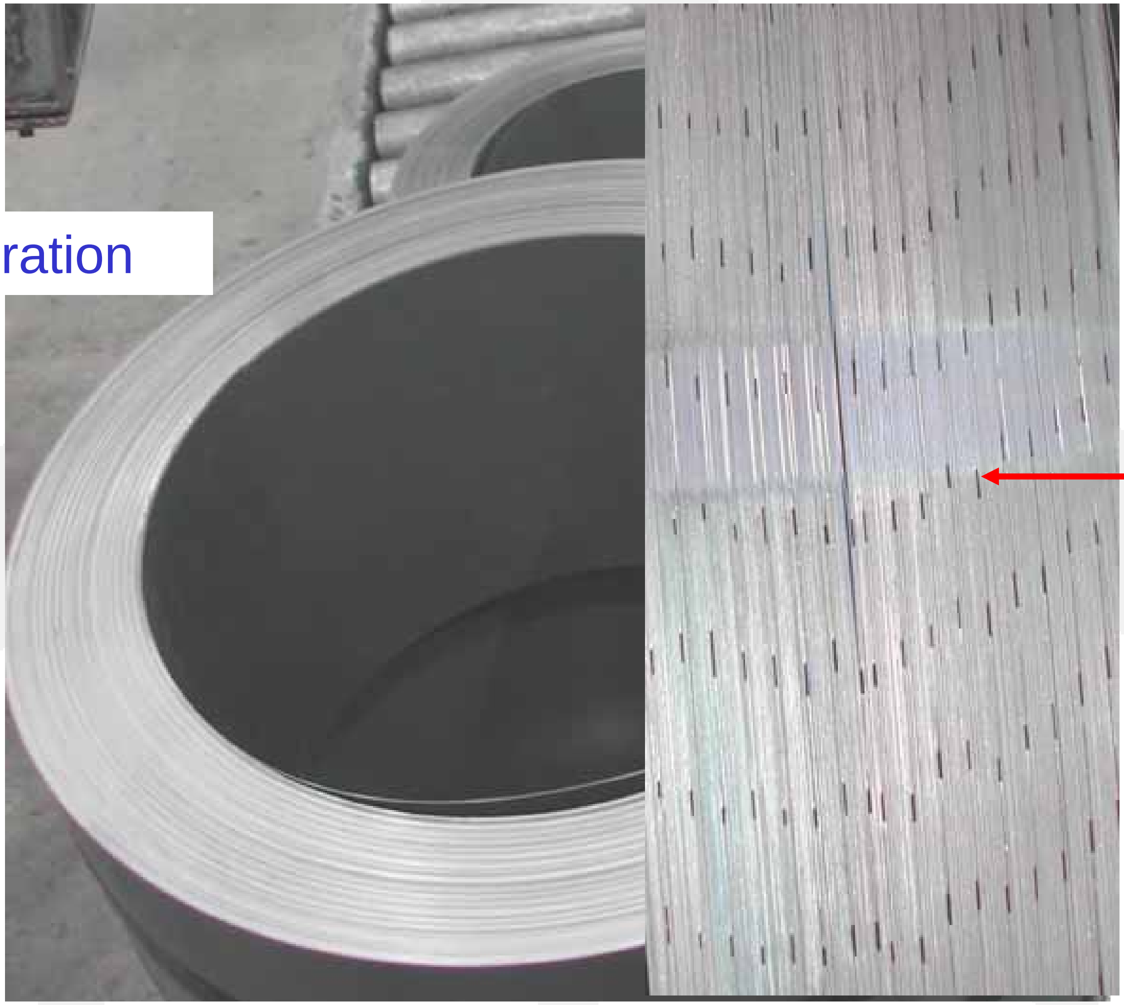
# Rectangular coil & 5 legged core assembly



# Distributed gap core construction

Core after shear operation

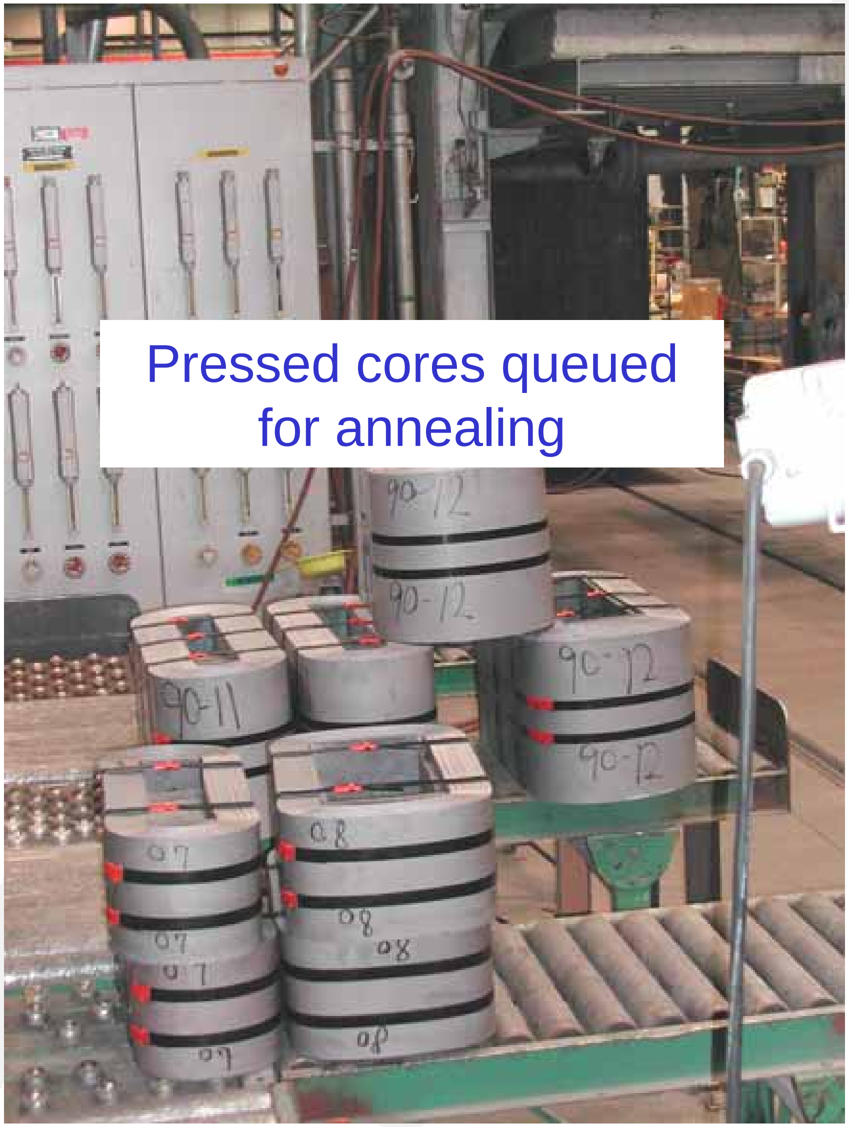


"Distributed" gaps

# Distributed gap core construction

Raw stock core material

# Distributed gap core construction




Pressed cores queued for annealing


Loading the annealing oven

# Distributed gap core construction

Annealed cores

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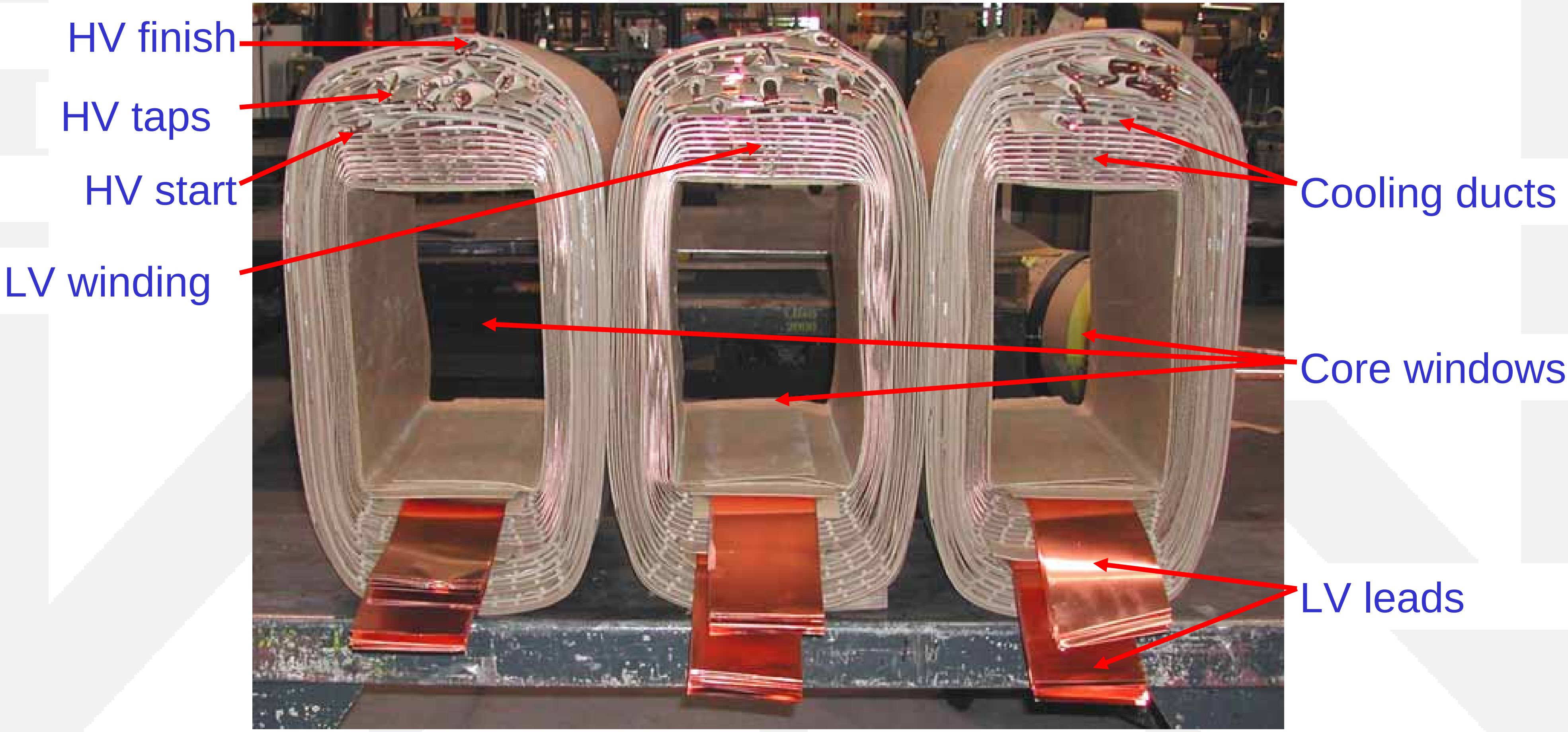
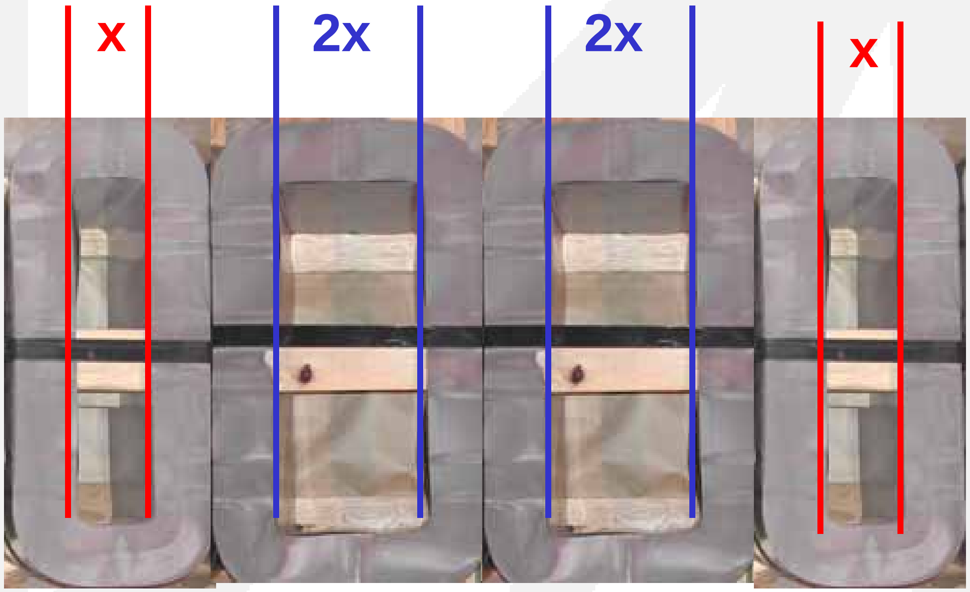
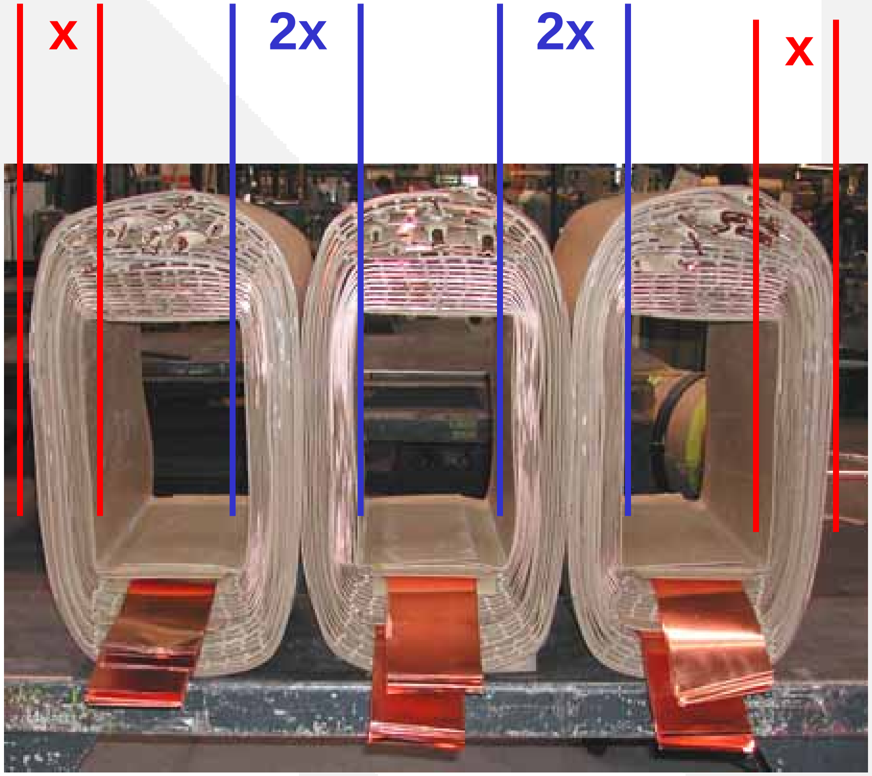
# Rectangular coils



# Rectangular coils w/ distributed gap cores



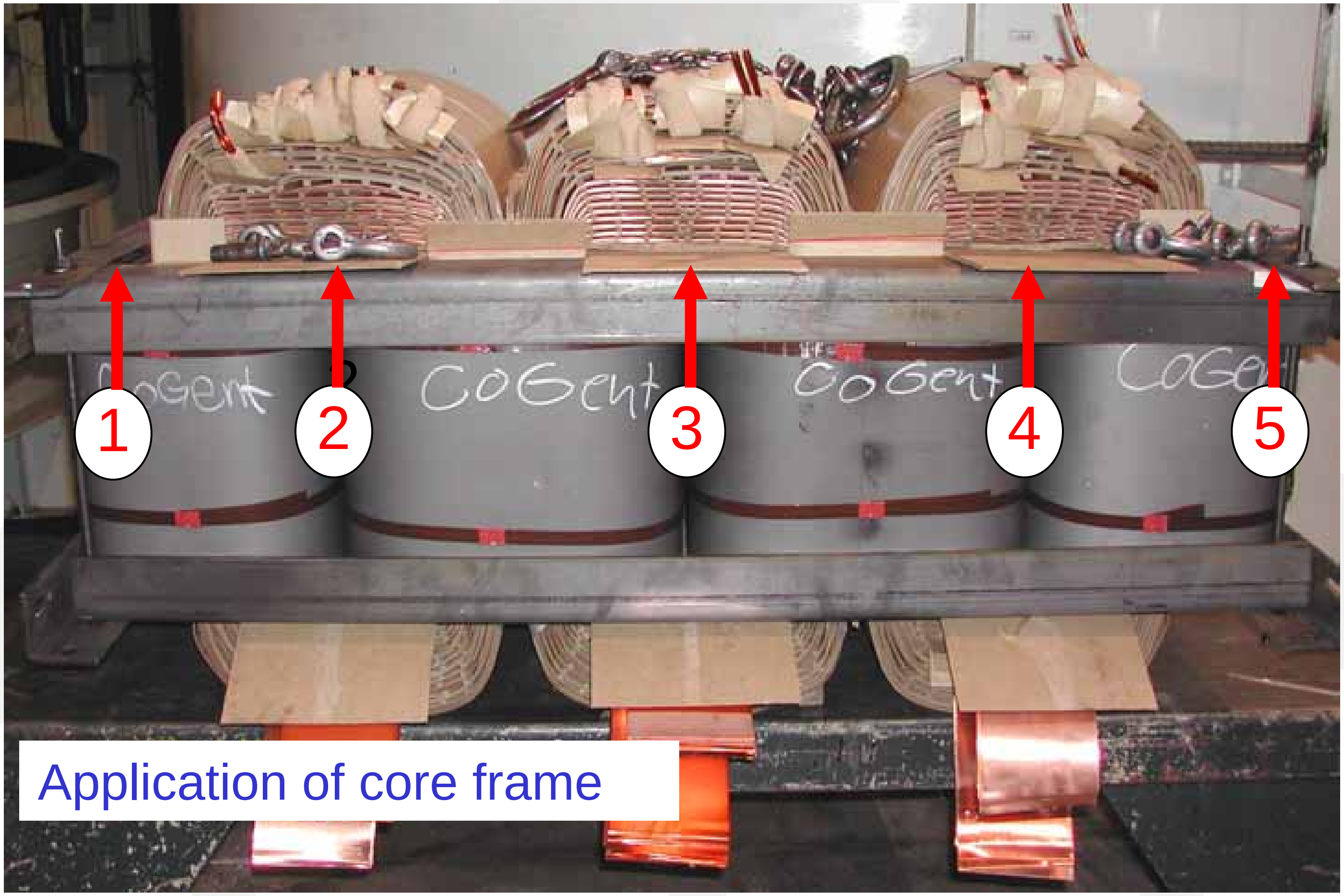



# Rectangular coils w/ distributed gap cores

Annealed Laminations

Stacking the distributed gap cores

# Rectangular coils w/ distributed gap cores



# Rectangular coils w/ distributed gap cores

Assembled 5 legged distributed gap core and coil assembly

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# Rectangular coils w/ distributed gap cores

Outer core legs and core frame

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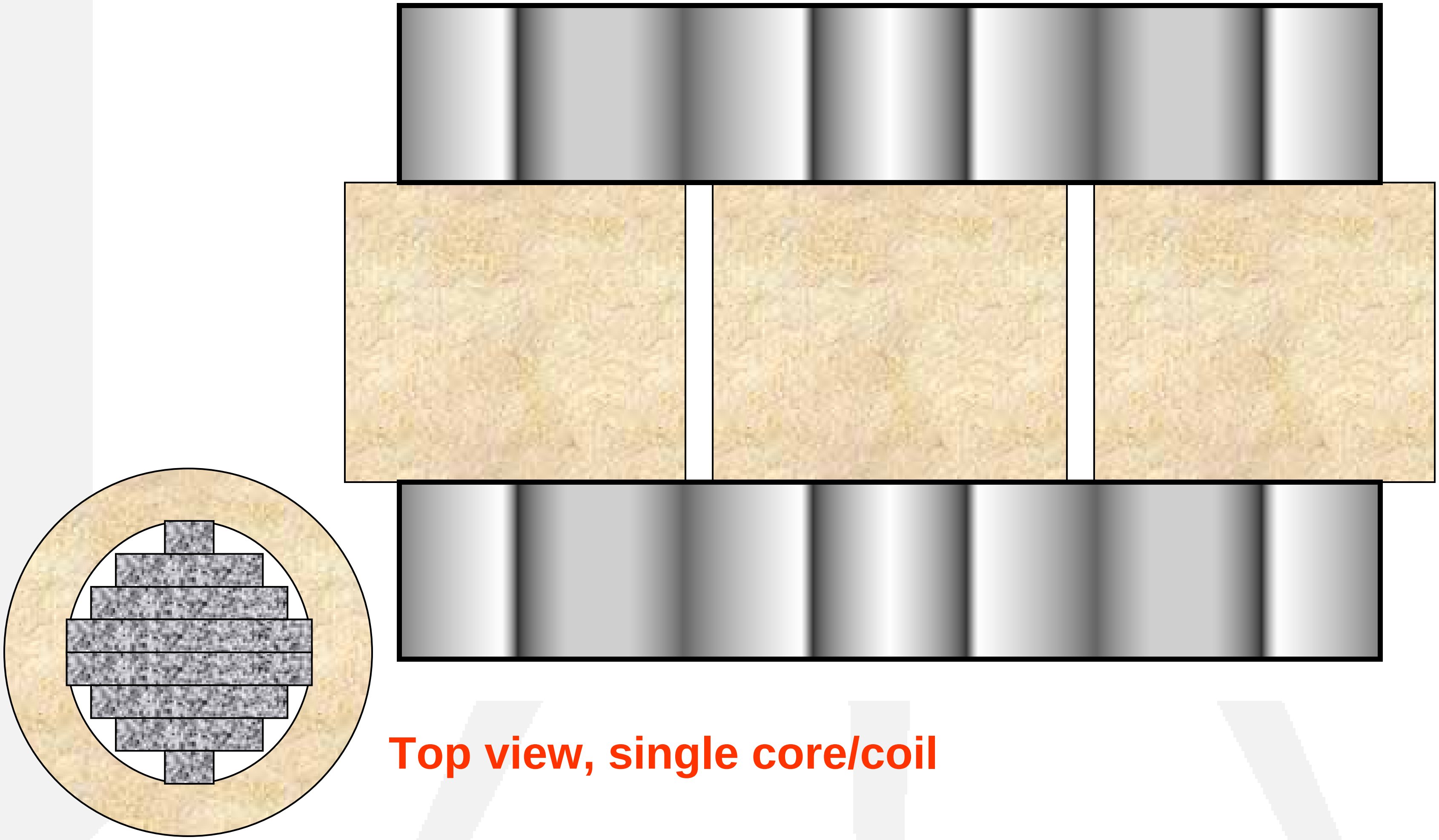
# Shell & core transformers

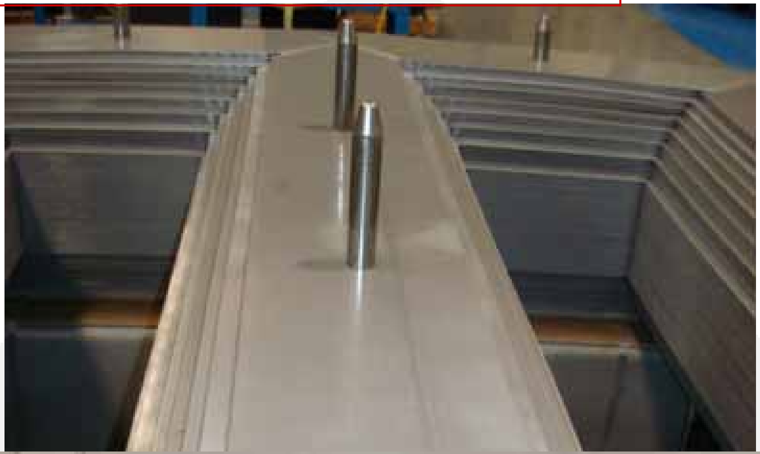
# Round core and coil assembly

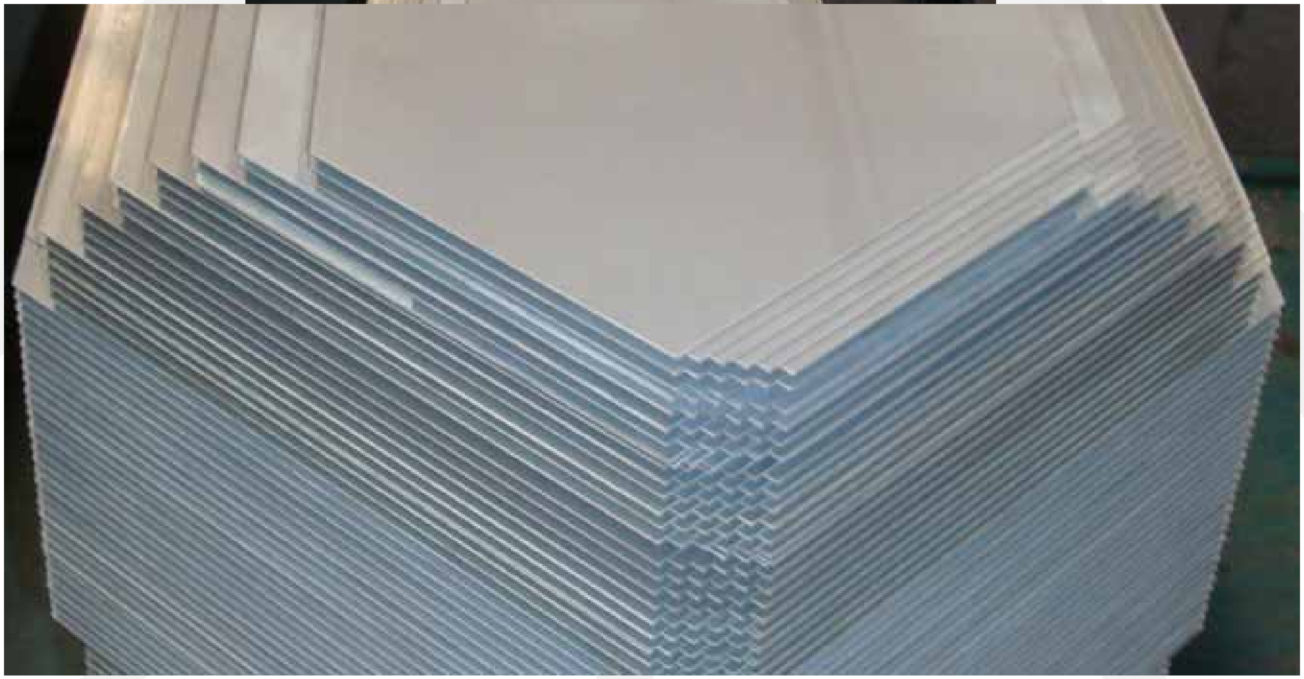


Top view, single core/coil

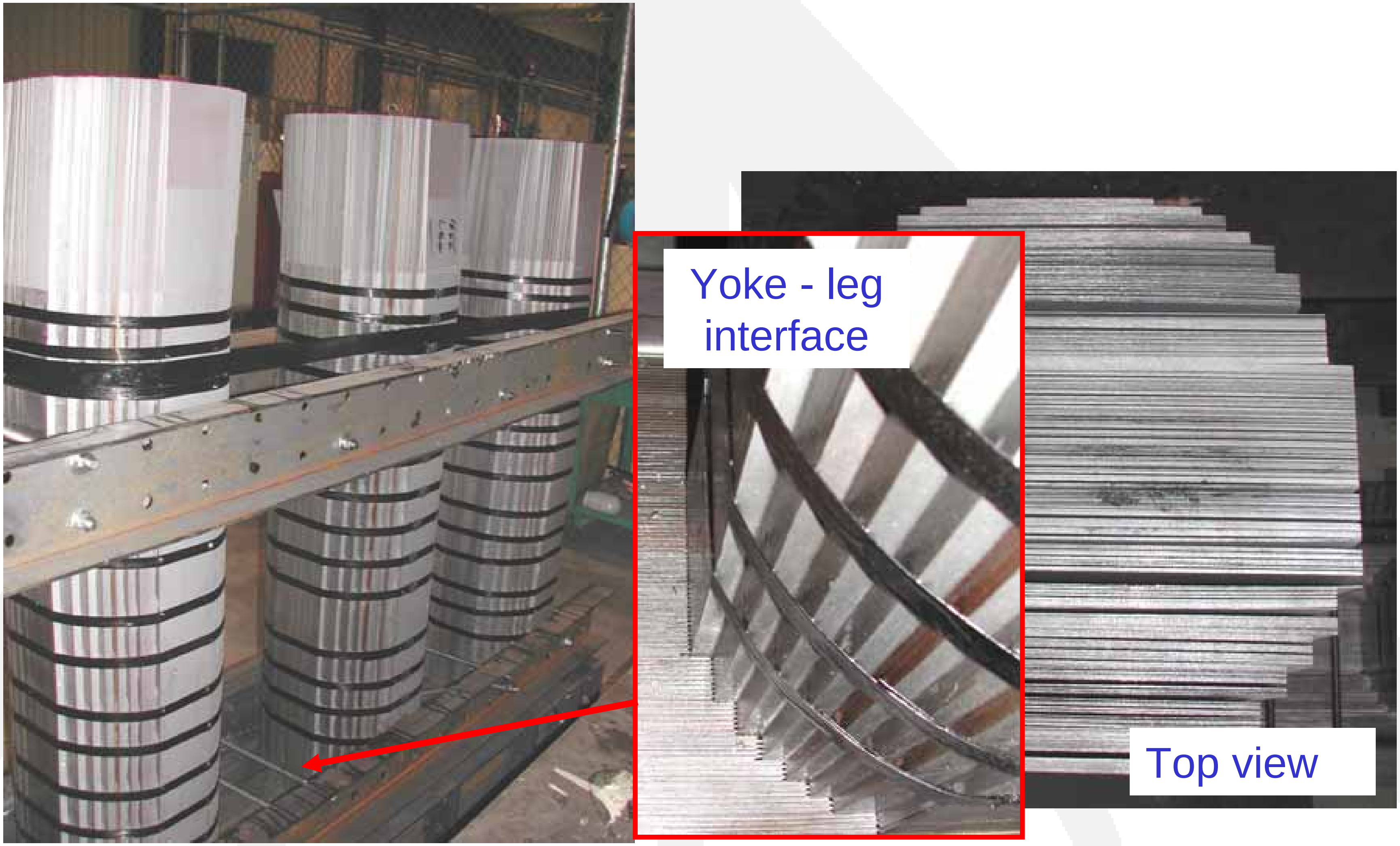
# Cruciform core shear

# Cruciform stacked core construction



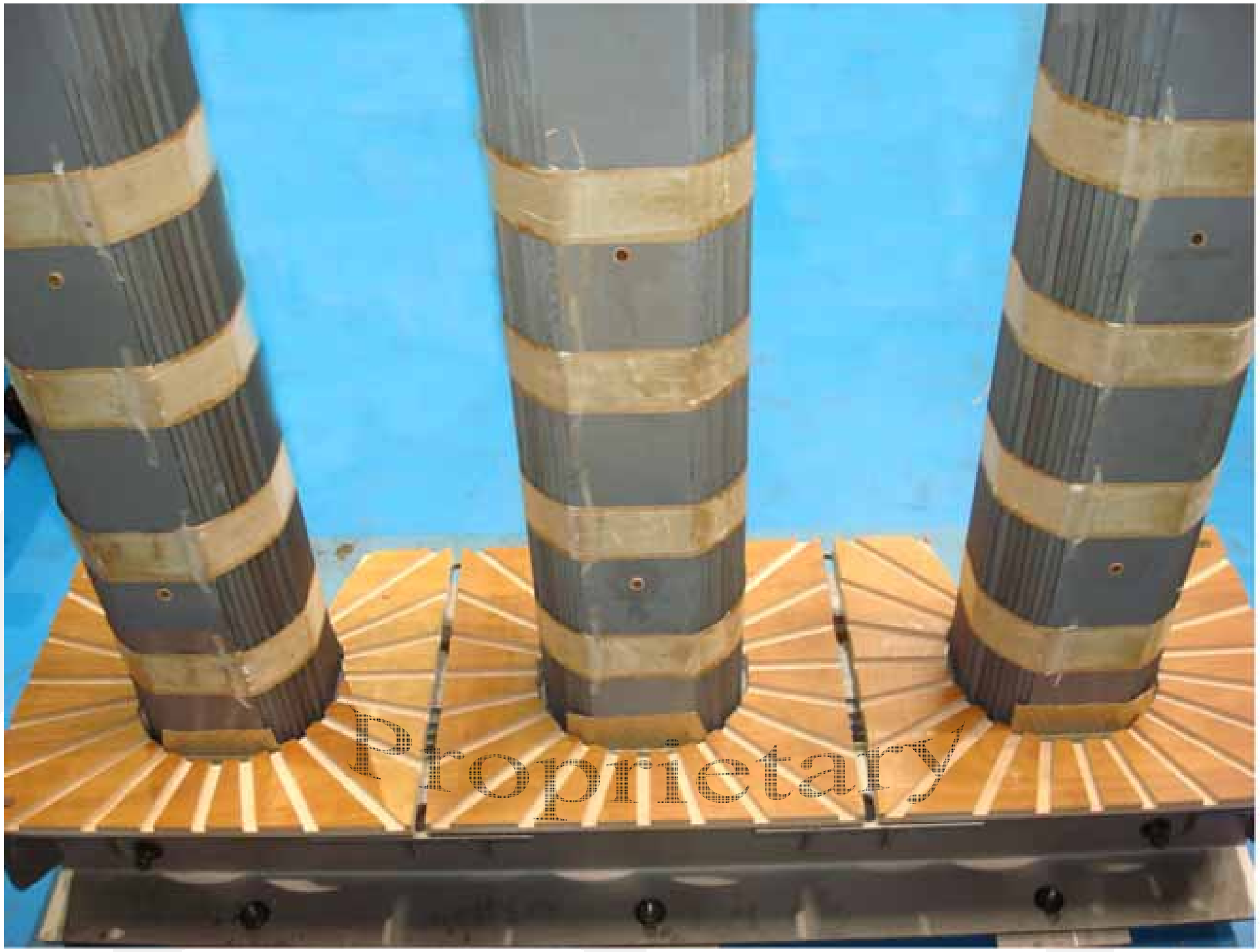


# Cruciform core construction



# Cruciform stacked core construction

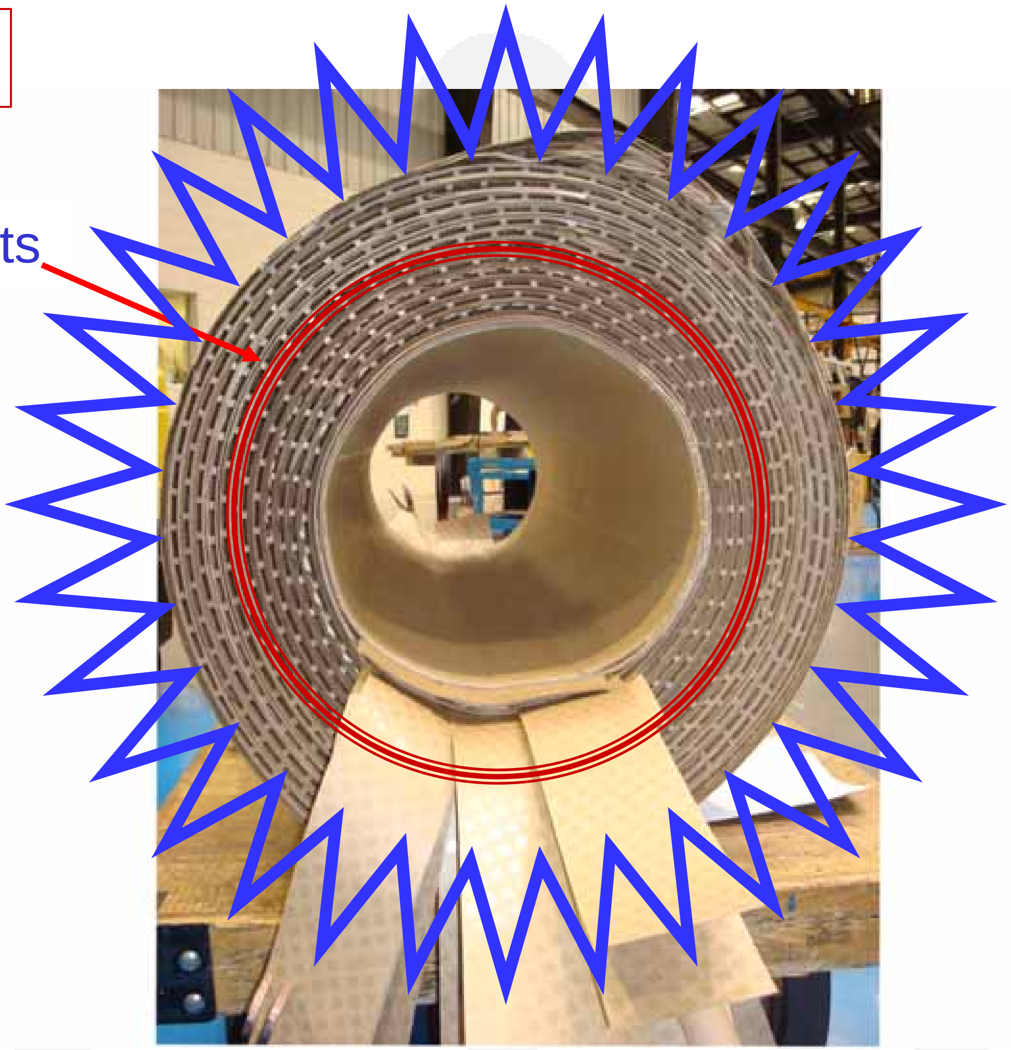
# Cruciform core



# Round Coils

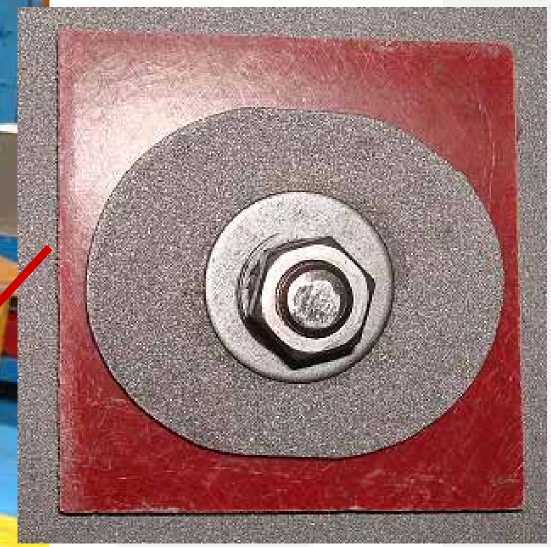
# Round Coils

360 degree cooling ducts



Radial forces are equalized during short circuits & overloads

# Core – Coil Assembly



Improved cooling

# Core – Coil Assembly

# Round coils with cruciform core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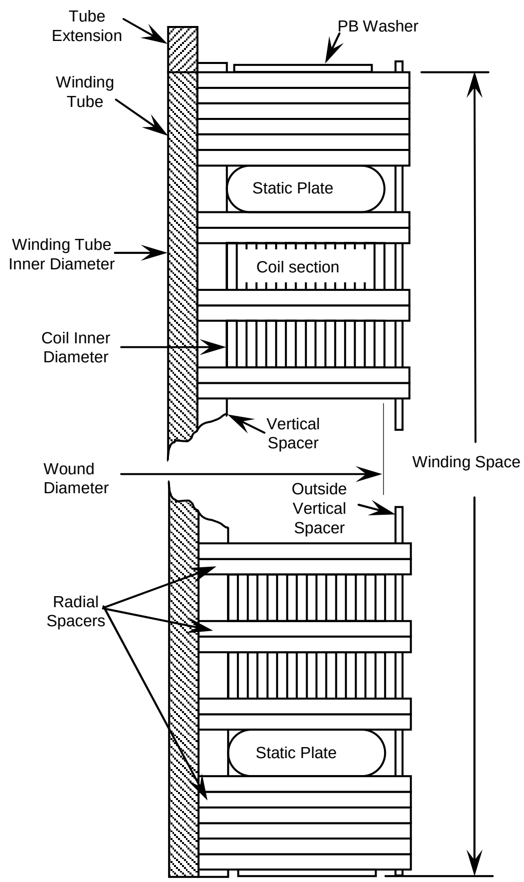
# Round coils with cruciform core

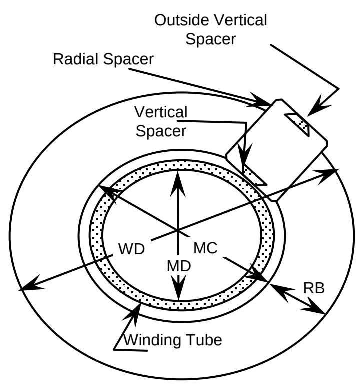
Tanking assembled core & coil assembly

## *Disk-Type Coil Winding*

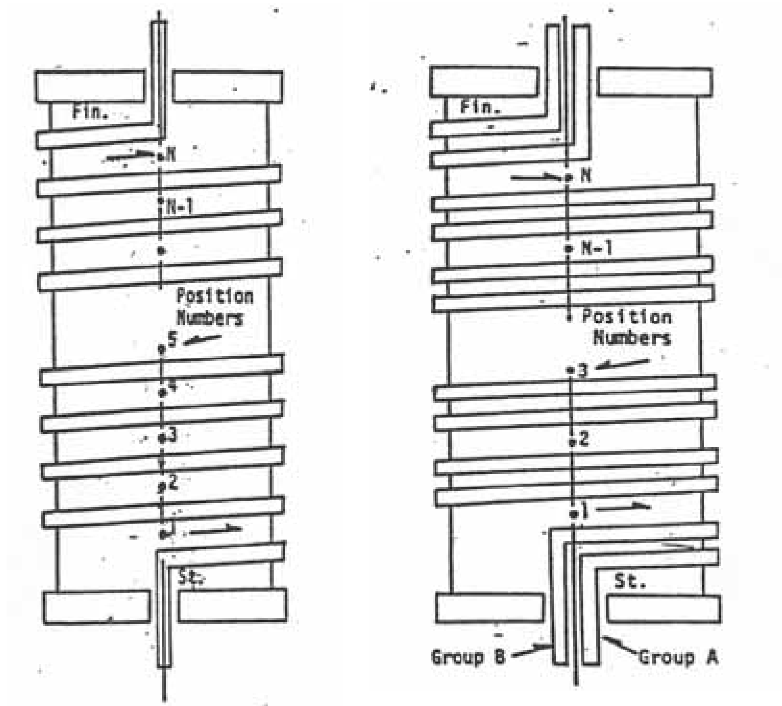
# Disk Coil Construction

## Continuous Disk




## Helical Disk



# Stages of Disk Windings Assembly

# Construction comparisons


## Core form VS. shell form

# Rectangular coil - 5 legged core advantages

With strip conductor secondaries:

- **HV electrically centered on the LV winding**
- **Significant gain in mechanical strength**
- **Suitable for high volume production**
- **No tank heating from stray flux induction in tank wall when connected wye-wye**

# Round coil - cruciform core uses

With wire / wire windings:

- **Means of providing increased hoop strength**
- **Enables solid construction of higher kVA units**
- **Suitable for small power designs & dedicated loads**
- **Reduction in width dimension for retrofit applications**

# Round Coil/Cruciform Core Advantages

- 360 degree cooling
  - Increased insulation life
  - More efficient cooling during heavy duty cycles
- Top & bottom pressure plates
  - Containment of axial forces during short circuits
- Round coil construction
  - Maximum containment of radial forces during short circuits
  - Improved clearances between phases

# Dielectric Fluids

- Mineral Oil (Type II)
- Vegetable-Based Oil (Envirotemp FR3)
  - High Fire Point (300 degrees C)
  - Biodegradable
- Silicone
- Voltesso
  - Lower Temperature Pour Point

# Standard Accessories

- Oil Temperature Gauge
- Liquid Level Gauge
- Pressure/Vacuum Gauge
- Pressure Relief Valve
- Drain Valve w/ Sample Port
- Upper Fill Port
- Air Space filled with Dry Nitrogen

# Optional Accessories Part 1

- Dry Contacts on Gauges
- Pressure Relief Device
- Forced Air Cooling
  - 15% increased capacity 2499 kVA and smaller
  - 25% increased capacity 2500 kVA and larger
- Winding Temperature Gauge
- Sudden Pressure Rise Relay
  - Seal-In Relay

# Optional Accessories Part 2

- Removable Radiators
- Positive Pressure System
- Stainless Steel
  - Auxiliary Wiring Cabinet – NEMA 4X
  - Hardware
  - Base
  - Tank
  - Radiators
  - Transitions

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# SUBSTATION SPECIFICATION SUMMARY SHEET

**Customer** ______________________ **Customer PO#** ______________ **SE:** ______

**CATALOG #:** ____________ **KVA:** ______ **QTY:** ______ **SO#:** ______ **PART#:** __________

**PHASE:**
- [ ] THREE
- [ ] SINGLE

**FLUID:**
- [ ] TYPE II OIL
- [ ] VOLT ESSO
- [ ] ENVIROTEMP(FR3)
- [ ] BIOTEMP
- [ ] SILICONE

**TEMP:**
- [ ] 65 C
- [ ] 55 C
- [ ] 55/65 C
- [ ] OTHER ______

**COOLING:**
- [ ] ONAN
- [ ] ONAN / ONAF-FFA
- [ ] ONAN / ONAF-less fans
- [ ] ONAN / ONAF

Note: If less flammable fluid is used, rating is KNAN or KNAF respectively.

**HERTZ:**
- [ ] 60 HZ
- [ ] 50 HZ

**CONDUCTOR:**
- # ALUMINUM
- # COPPER

**AMBIENT:**
- [ ] 30 C avg.
- [ ] 40 C avg.
- [ ] OTHER ______

- [ ] LTC
- [ ] GROUNDING
- [ ] RECTIFIER: ____ PULSE

- [ ] 5 LEGGED CORE
- [ ] TERTIARY
- [ ] TRIPLEX CORE/COIL

**HIGH VOLTAGE**

**HV:** ____________________

**BIL:** ______ **kV**

**CONNECTION:**
- [ ] DELTA
- [ ] WYE
  - [ ] HO Bushing w/full BIL
  - [ ] HO Ground Strap
  - [ ] Neutral internal & isolated
  - [ ] Neutral internally grounded
- [ ] GROUNDED WYE (HO-XO Comm.)
- [ ] DELTA-WYE
- [ ] ZIG ZAG

**TAPS:**
- [ ] 2-2 1/2% FCAN & BN
- [ ] 4-2 1/2% FCBN
- [ ] 2-2 1/2% FCAN & 4-2 1/2% FCBN
- [ ] OTHER ____________

- [ ] ES SHIELD
- [ ] BALANCED TAPS
- [ ] WET WIND

**IMPEDANCE:** ______ %
- [ ] MIN
- [ ] NOM
- [ ] MAX

**VECTOR CONNECTION:** ______

**LOW VOLTAGE**

**LV:** ____________________

**BIL:** ______ **kV**

**CONNECTION:**
- [ ] DELTA
  - [ ] With single Phase Mid-Tap Coil
- [ ] WYE
  - [ ] XO bushing
  - [ ] XO bushing w/ground strap
- [ ] GROUNDED WYE (XO-HO Comm.)
- [ ] DELTA-WYE
- [ ] ZIG ZAG

**HV LOCATION:** [ ] Left [ ] Right [ ] Tank face Seg ______ [ ] Top Seg ______

**LV LOCATION:** [ ] Left [ ] Right [ ] Tank face Seg ______ [ ] Top Seg ______

**HV BUSHINGS:** **HV CONNECTION:** **LV BUSHINGS:** **LV CONNECTION:**

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Thank you!